

TOPIC NAME: Fall 2025 (A)

DAY: _____

Azmari Sultana

TIME: _____

DATE: / /

① b)

1 main:

2 int p=2, q=3, n=4

3 pid1 = fork()

4 if (pid1 == 0):

5 p = p + 3

6 q = q + 1

7 n = n + 2

8 printf("p:", p, "q:", q, "n:", n)

9 else:

10 wait(NULL)

11 p = p + 4

12 q = q + 5

13 n = n + 6

14 pid2 = fork()

15 if (pid2 == 0):

16 p = p + 1

17 q = q + 1

18 n = n + 1

18 printf("p:", p, "q:", q, "n:", n)

19 else:

20 wait(NULL)

21 printf("p:", p, "q:", q, "n:", n)

22

P1
p=5, q=4, n=6
pid1=0
pid2=0

P2
p=6, q=8, n=10
pid1=0
pid2=0

P3
p=7, q=9, n=11
pid1=0
pid2=0

Output:

p: 5, q: 4, n: 6 → P2

p: 7, q: 9, n: 11 → P3

p: 6, q: 8, n: 10 → P1

GOOD LUCK™

TOPIC NAME: Spring 25 (A)

DAY: _____

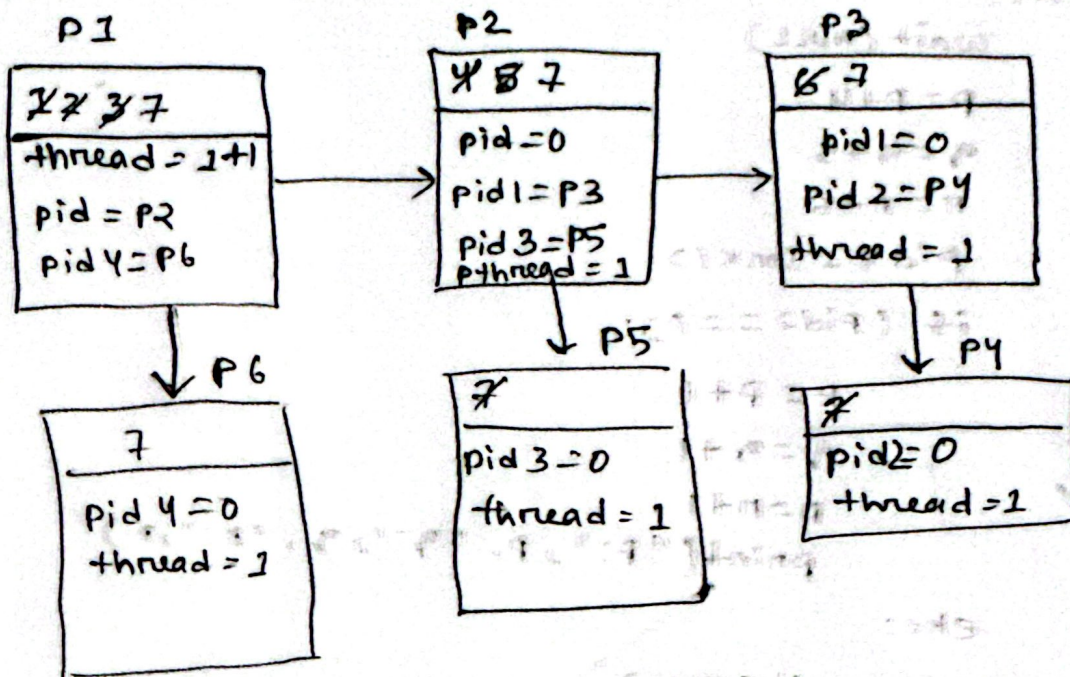
TIME: _____

DATE: / /

① ii)

```

1 pid = &pid;
2 pthread_create(...);
3 pid = fork();
4 if (pid == 0) { /* child process */
5     fork();
6 }
7 fork();
    
```



Thread = 7

Process = 6

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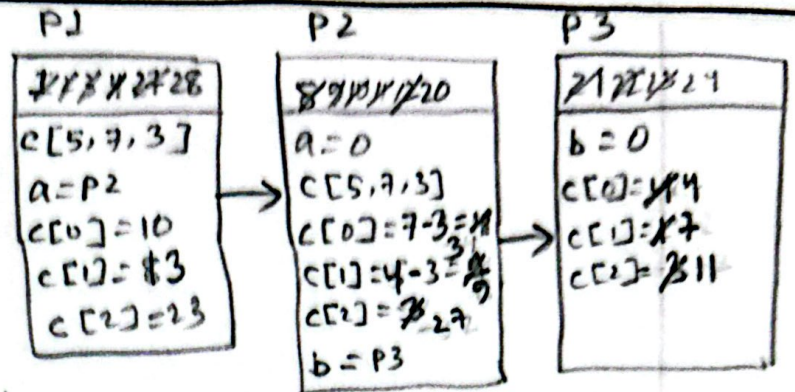
DATE: / /

b)

```

1 int main() {
2     pid_t a, b;
3     int c[] = {5, 7, 3};
4     a = fork();
5     if (a < 0) {
6         printf("error\n");
7     }
8     else if (a == 0) {
9         c[0] = c[1] - c[2];
10        c[1] = c[0] - c[2];
11        c[2] = c[0] - c[1];
12        b = fork();
13        if (b < 0) {
14            }
15        else if (b > 0) {
16            wait();
17            c[0] = c[1] + c[2];
18            c[1] = c[0] * c[2];
19            c[2] = c[0] * c[1];
20        }
21        else {
22            c[0] = c[1] + c[2];
23            c[1] = c[0] + c[2];
24            c[2] = c[0] + c[1];
25        }
26    }
27    else {
28        printf("hello\n");
29        wait();
30        c[0] = c[1] + c[2];
31        c[1] = c[0] + c[2];
32        c[2] = c[0] + c[1];
33    }
}

```



Output:

```

hello → P1
c[0] = 10
c[1] = 13
c[2] = 23 } P1

c[0] = 3
c[1] = 9
c[2] = 27 } P2

c[0] = 4
c[1] = 1
c[2] = 11 } P3

```

```

34 for (int i = 0; i < 3; i++) {
35     printf("c[%d] = %d\n", i, c[i]);
36 }
37 return 0;
}

```

① b)

1 main() {

2 a = fork();

3 b = fork();

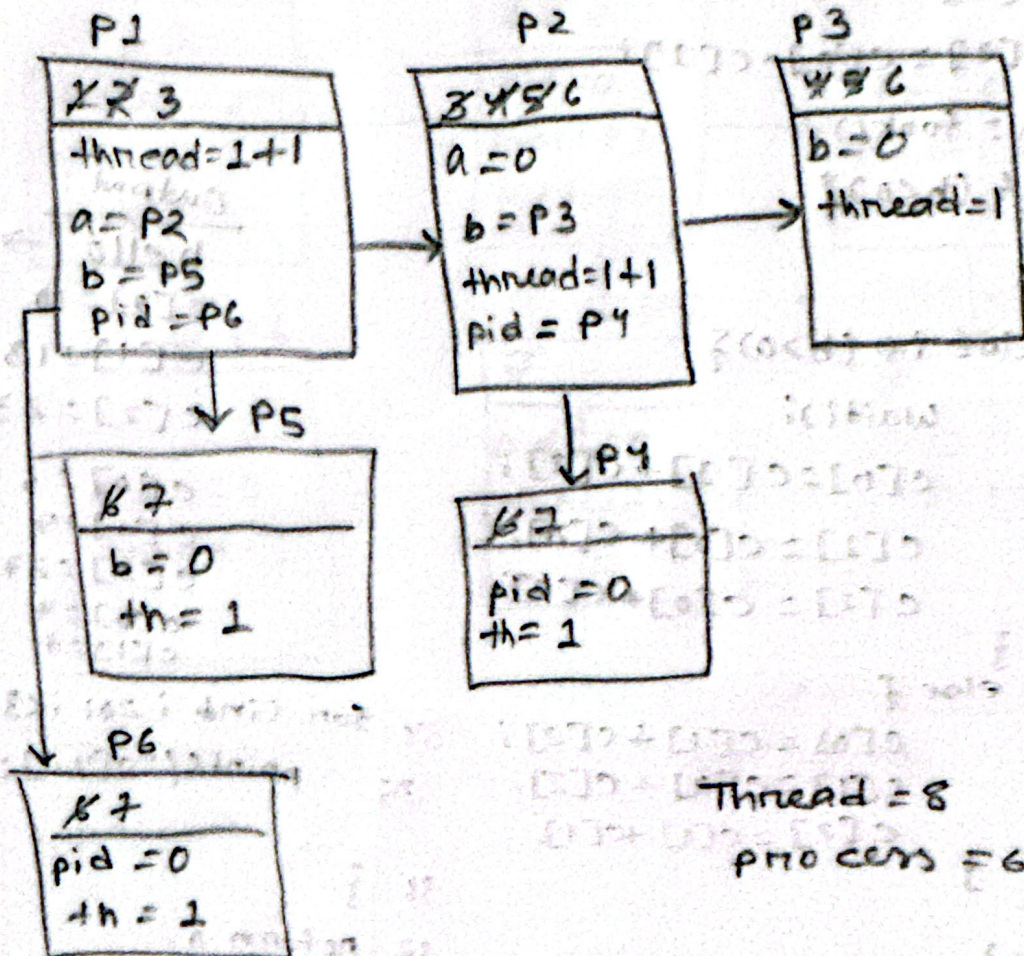
4 if (b > 0) {

5 pthread_create(...);

6 fork();

 }

}



TOPIC NAME: _____

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d)

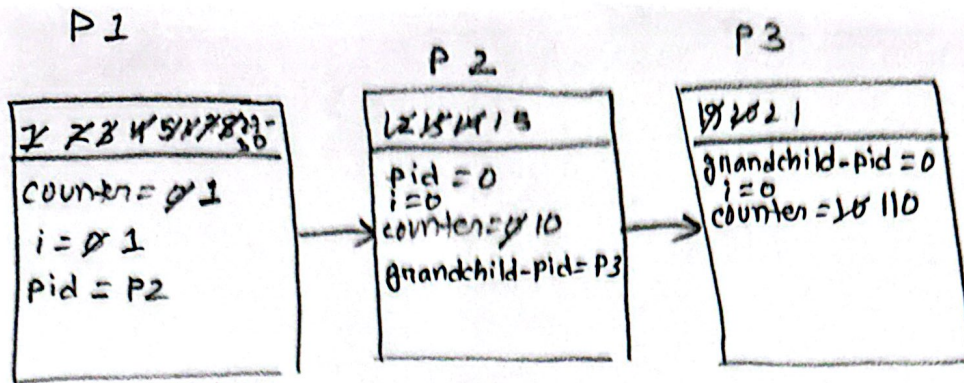
```

1 int main() {
2     int counter=0;
3     int i;
4     printf("Initial process starting. Counter: %d\n", counter);
5     for (i=0; i<2; i++) {
6         if (i==0) {
7             pid_t pid = fork();
8             if (pid < 0) {
9                 printf("Error\n");
10                return 1;
11            } else if (pid == 0) {
12                counter += 10;
13                printf("Child from loop %d: its counter is now %d.\n", i, counter);
14                pid_t grandchild_pid = fork();
15                if (grandchild_pid < 0) {
16                    printf("Error forking grandchild.\n");
17                    return 1;
18                } else if (grandchild_pid == 0) {
19                    counter += 100;
20                    printf("Grandchild from loop %d: its counter is\n", i, counter);
21                    now %d.\n", i, counter);
22                } else {
23                    wait(NULL);
24                    counter += 1;
25                }
26                break;
27            } else {
28                wait(NULL);
29                counter += 1;
30                printf("Parent after loop %d: its counter is now %d.\n", i, counter);
31            }
32        } else {
33            printf("Process in 2nd loop iteration %d: its counter is %d.\n", i, counter);
34            break;
35        }
36    }
37    return 0;
}

```

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output

Initial process starting. counter: 0
→ P1

Child from loop 0: its counter is now 10
→ P2

Grandchild from loop 0: its counter is now 110
→ P3

parent after loop 0: its counter is now 1
→ P1

process in second loop iteration 1:

its counter is 1.
→ P1